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BanglaSarcThematic: A Transformer-Based Approach towards Fine-Grained Thematic Sarcasm Analysis Using a Novel Dataset

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***Abstract*—Sarcasm detection in Bengali texts presents unique challenges due to complex linguistic structures and cultural nuances. While existing approaches focus on binary classification (sarcasm or non-sarcasm), this paper introduces BanglaSarcThematic, the first transformer-based framework for fine-grained thematic sarcasm analysis in Bengali. We contribute a comprehensive dataset of 12,410 manually annotated Bengali social media comments with five labels: non-sarcasm and four thematic sarcasm categories (non-thematic, geopolitical, hate expression, and entertainment sarcasm). Our methodology evaluates two classification strategies using six state-of-the-art transformer models: BERT-based models for bidirectional contextual understanding and contrastive models for discriminative semantic learning. We compare direct five-class classification against hierarchical two-stage classification. Our hierarchical approach demonstrates effectiveness through specialized model selection: Stage 1 achieves 84.21% binary classification(sarcasm or non-sarcasm) accuracy using BanglaBERT, while Stage 2 achieves 88.75% accuracy on thematic classification of sarcastic content using BGE-M3 Embedding. The end-to-end pipeline yields 77.13% overall accuracy compared to 76.15% for direct five-class classification. The superior performance of specialized models at different stages validates the effectiveness of hierarchical classification for capturing thematic nuances in Bengali sarcastic discourse, establishing new benchmarks for low-resource language sarcasm detection.**

***Index Terms*—sarcasm detection, Bengali language processing, transformer models, thematic sarcasm classification, natural language processing, BGE-M3 Embedding**

I. Introduction

Sarcasm detection has emerged as a critical challenge in natural language processing, particularly for low-resource languages like Bengali. The ability to identify and categorize sarcastic expressions is essential for various applications, including sentiment analysis, social media monitoring, and automated content moderation. While traditional research has focused primarily on binary sarcasm detection, the complexity of sarcastic expressions necessitates more nuanced analysis that considers thematic contexts and cultural specificities.

Bengali, spoken by over 265 million people worldwide, presents unique computational challenges due to its rich morphological structure, complex syntax, and culturally embedded linguistic expressions. Sarcastic content in Bengali often relies heavily on contextual understanding and cultural knowledge, making it particularly challenging for automated systems to detect and classify accurately. Moreover, the diversity of sarcastic expressions across different domains—political commentary, entertainment criticism, social issues—requires fine-grained categorization beyond simple binary classification.

Existing Bengali sarcasm detection systems predominantly focus on binary classification, failing to capture the thematic richness inherent in sarcastic discourse. This limitation hampers applications requiring a nuanced understanding of sarcastic intent, such as targeted content moderation, political sentiment analysis, and cultural trend monitoring. The lack of thematically annotated datasets further constrains research advancement in this domain.

This paper introduces BanglaSarcThematic, a novel transformer-based framework that addresses the gap in fine-grained thematic sarcasm analysis for Bengali texts. Unlike existing binary classification systems, our approach not only distinguishes sarcastic from non-sarcastic content, but it also categorizes sarcastic instances into four thematic dimensions: non-thematic sarcasm, geopolitical sarcasm, sarcastic hate expression, and entertainment sarcasm.

Key contributions of our work include:

- First thematic sarcasm classification framework for Bengali, categorizing sarcastic content into four distinct themes beyond binary classification.
- Novel thematically-annotated Bengali sarcasm dataset with 12,410 manually annotated social media comments, the largest of its kind for Bengali sarcasm research.

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- Empirical demonstration that hierarchical task decomposition with specialized models outperforms direct classification for Bengali thematic sarcasm detection.
- Comprehensive transformer model evaluation across six state-of-the-art models, identifying optimal combinations: BanglaBERT for binary detection and BGE-M3 Embedding for thematic classification.

II. Related Work

Sarcasm detection has been extensively studied for high-resource languages, with research spanning traditional machine learning approaches using handcrafted features to modern deep learning architectures. Early approaches relied on linguistic features, syntactic patterns, and contextual cues [1], while recent developments have leveraged transformer architectures and pre-trained language models to achieve state-of-the-art performance in binary sarcasm classification tasks [2].

Despite Bengali being the seventh most spoken language worldwide with over 265 million speakers, sarcasm detection research in Bengali remains significantly limited compared to high-resource languages like English. The scarcity of annotated datasets and computational resources has hindered comprehensive research in this domain, with only a handful of studies addressing Bengali sarcasm detection in recent years.

Lora et al. [3] introduced Ben-Sarc, a self-annotated corpus for Bengali sarcasm detection containing social media comments. Their work achieved 75.05% accuracy using Indic-Transformers Bengali BERT as a pre-trained model.

The BanglaSarc dataset [4] contributed 5,112 comments collected from various online social platforms, providing a foundational resource for Bengali sarcasm research. However, this dataset focused primarily on binary classification without thematic categorization.

Biswas et al. [5] proposed a hybrid RNN-BiLSTM model leveraging bidirectional context for morphologically rich Bengali syntax, achieving 89.6% accuracy with 5.12-16.82% improvement over prior work. Their ternary classification approach reduced ambiguity-driven errors by 18% compared to binary frameworks, highlighting the importance of refined classification granularity.

Pal and Prasad [6] developed a neural network approach combining sarcasm detection with sentiment analysis for Bengali. Using GloVe word representations and LSTM, they achieved 91.94% accuracy.

Khanom et al. [7] conducted a comparative analysis using multiple AI approaches on 5,000 Bengali texts. Their fine-tuned BanglaBERT achieved superior performance with 83% accuracy and 82.55% F1-score.

Recent dataset contributions include BanglaSarc3 [8], providing a ternary-class dataset with 12,089 Facebook comments categorized as neutral, sarcastic, and non-sarcastic. Muntaha et al. [9] explored explainable sarcasm detection using Bi-LSTM with LIME explanations, achieving 72.10% accuracy while providing model interpretability for Bengali sarcasm detection.

Research Gap and Contribution Positioning: Despite the growing interest in Bengali sarcasm detection, existing research exhibits critical limitations. All previous work focuses exclusively on binary or ternary classification, ignoring the rich thematic diversity inherent in sarcastic expressions. Dataset sizes remain modest (5,000-12,000 samples), and no prior work addresses cultural and contextual categorization essential for comprehensive sarcasm understanding in Bengali digital discourse.

Our work addresses these fundamental gaps by introducing the first thematic sarcasm analysis framework for Bengali. Unlike previous binary approaches, we provide fine-grained categorization across four thematic dimensions, supported by the largest thematically-annotated Bengali sarcasm dataset (12,410 comments). Additionally, we present the first comprehensive transformer model comparison specifically designed for Bengali thematic sarcasm tasks, establishing new benchmarks for this underexplored domain.

III. Data Overview

A. Dataset Construction and Collection

Our study introduces a comprehensive dataset of 12,410 manually annotated Bengali comments specifically designed for fine-grained thematic sarcasm analysis. The dataset was systematically collected from diverse social media platforms, including Facebook, X (formerly known as Twitter), YouTube, and Instagram, focusing on user-generated content that exhibits authentic patterns of Bengali sarcastic expression in contemporary digital communication contexts.

The data collection strategy employed stratified sampling across different content domains to ensure representative coverage of Bengali sarcastic discourse. Comments and posts were selected based on engagement metrics and linguistic diversity, prioritizing content that demonstrates cultural specificity and contextual richness characteristic of Bengali social media communication.

B. Annotation Framework and Thematic Categories

The annotation scheme follows a hierarchical two-level framework designed to capture both binary sarcasm identification and fine-grained thematic categorization. The primary level distinguishes sarcastic from non-sarcastic content, while the secondary level categorizes sarcastic instances into four distinct thematic dimensions:

- **Non-thematic Sarcasm**: General sarcastic expressions lacking specific domain focus or cultural targeting.
- **Geopolitical Sarcasm**: Expressions addressing national and international political events, government policies, electoral processes, or socio-political commentary
- **Sarcastic Hate Expression**: Content targeting sensitive topics, including religious, ethnic, gender, or caste-based issues with sarcastic undertones.
- **Entertainment Sarcasm**: Content directed toward sports events, the entertainment industry, celebrities, or cultural activities.

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C. Dataset Distribution and Statistics

Table I presents the comprehensive distribution of the dataset across all categories, demonstrating balanced representation across thematic dimensions while reflecting natural occurrence patterns in Bengali social media discourse.

Classes	Sub-classes	Word Count	Samples
Non-Sarcasm	-	87,646	6,192
Sarcasm	Non-thematic	42,973	3,162
	Geopolitical	9,494	807
	Hate expression	12,213	904
	Entertainment	22,403	1,345
Total		**174,729**	**12,410**

TABLE I
DATASET DISTRIBUTION ACROSS CLASSES AND THEMATIC CATEGORIES

D. Dataset Examples and Category Illustration

Figure 1 presents representative examples demonstrating the linguistic diversity and cultural specificity of Bengali sarcastic discourse across our thematic framework. The examples illustrate distinct patterns: non-sarcastic content exhibits straightforward communication, while sarcastic categories demonstrate varying cultural contexts and ironic structures essential for automated classification.

Comments	Categories
ধন্যবাদ তোমার সাহায্যের জন্য!	Non-Sarcasm
আরেকটু পরেই আসতে অনুষ্ঠান তো প্রায় শেষ।	Non-thematic Sarcasm
রোহিঙ্গা সমস্যা তো কোনো সমস্যাই না, ওরা তো আমাদের মেহমান	Geopolitical Sarcasm
দলটা এত ধারাবাহিক যে প্রত্যেক ম্যাচেই একইভাবে হারে।	Entertainment Sarcasm
বস্তির মেয়ে সুমাইয়ার জন্য কেয়া কসমেটিকস থেকে আকর্ষণীয় পুরষ্কার	Sarcastic Hate Expression

Fig. 1. Representative examples from each category showing linguistic patterns and thematic variations

E. Annotation Quality and Inter-Annotator Agreement

Annotation quality was ensured through rigorous validation procedures involving four native Bengali speakers with linguistic expertise. Each annotator underwent comprehensive training on annotation guidelines, cultural context recognition, and thematic categorization criteria. The annotation process followed a two-phase approach: initial independent annotation followed by consensus-building discussions for disputed cases.

Inter-annotator agreement was measured using Fleiss' Kappa coefficient [10], which accounts for chance agreement among multiple raters. The coefficient is calculated as:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

where $\bar{P}$ represents the proportion of observed agreement and $\bar{P}_e$ represents the proportion of agreement expected by chance. Our annotation process achieved a Fleiss' Kappa of 0.82, indicating substantial agreement according to Landis and Koch's interpretation guidelines [11], confirming the reliability and consistency of our thematic categorization framework.

F. Data Partitioning

The dataset was partitioned using stratified random sampling with an 80:10:10 split ratio, ensuring balanced representation of all classes across training (9,928 samples), validation (1,241 samples), and test (1,241 samples) sets. For hierarchical classification, Stage 2 operates exclusively on sarcastic instances distributed across four thematic categories as shown in Table I. This partitioning strategy maintains proportional class distribution across all splits while enabling specialized thematic analysis in the hierarchical framework.

IV. METHODOLOGY

A. Data Preprocessing

Input text underwent standard preprocessing, including Unicode normalization for consistent Bengali script representation and removal of URLs and emojis while preserving semantic content. Each transformer model employed its respective pretrained tokenizer, with BanglaBERT utilizing the specialized Bengali normalizer developed by Hasan et al. [12] for optimized Bengali text normalization.

B. Classification Approaches

Our methodology evaluates two distinct classification paradigms using six state-of-the-art transformer models, designed to investigate the effectiveness of specialized versus unified approaches for thematic sarcasm detection.

Direct Five-Class Classification: This approach performs simultaneous classification of input text into five categories: non-sarcastic, non-thematic sarcasm, geopolitical sarcasm, hate expression sarcasm, and entertainment sarcasm. The model processes Bengali text through a single transformer encoder followed by a five-way classification head with softmax activation.

Figure 2 illustrates our five-class direct classification architecture.

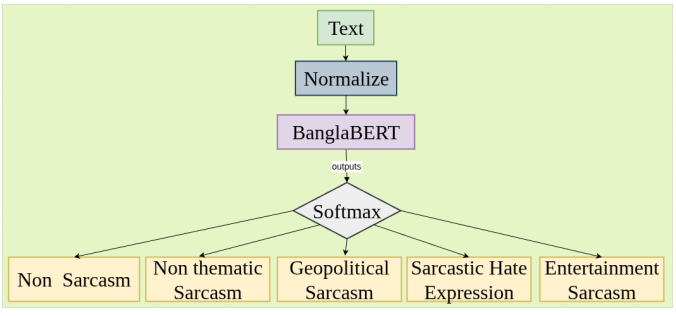


Fig. 2. Five-class direct classification architecture

Hierarchical Two-Stage Classification: This approach employs a cascaded architecture where Stage 1 performs binary sarcasm detection on the complete dataset, followed by Stage

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2, which applies thematic categorization exclusively to sarcastic instances.

We evaluate this approach using two methodologies:

- **Standalone Performance:** Stage 2 models are trained and evaluated on sarcastic instances to assess thematic classification capability independent of Stage 1 errors.
- **End-to-End Pipeline Performance:** Stage 2 is applied only to instances predicted as sarcastic by Stage 1, providing realistic system performance with error propagation.

Figure 3 illustrates our proposed hierarchical framework.

After individual evaluation, we selected BanglaBERT for Stage 1 and BGE-M3 Embedding for Stage 2 to construct the final pipeline. The overall pipeline accuracy represents the end-to-end system performance, where Stage 2 processes only Stage 1 sarcastic predictions.

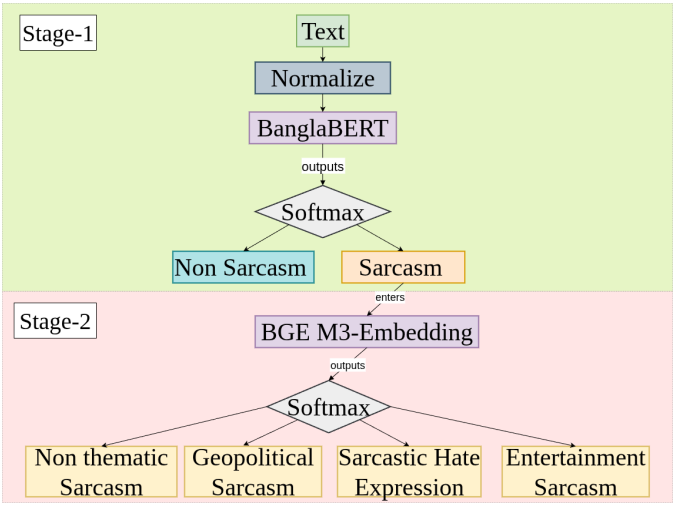


Fig. 3. Hierarchical Two-Stage Classification Framework architecture

C. Transformer Models

1) Traditional BERT-based Models: BERT-based models leverage bidirectional encoder representations with masked language modeling pre-training, providing strong contextual understanding capabilities:

- **Bangla BERT(base)**: Bengali-specific BERT model (110M parameters) pre-trained on large-scale Bengali corpora [13], optimized for the Bengali language.
- **BanglaBERT**: Enhanced Bengali BERT variant [14] (110M parameters) with improved training on diverse text sources and specialized normalization [12], specifically designed for Bengali language processing tasks
- **XLM-RoBERTa**: Cross-lingual RoBERTa model (279M parameters) trained on 100 languages using optimized training procedures [15].

2) Contrastive Models: Contrastive models leverage contrastive learning techniques to learn discriminative representations by optimizing similarity between semantically related instances while pushing apart dissimilar ones:

- **BGE-M3 Embedding**: State-of-the-art multilingual contrastive model (568M parameters) trained using contrastive learning for cross-lingual semantic understanding [16], with particular strength in capturing subtle semantic differences.
- **paraphrase-multilingual-MiniLM-L12-v2**: Compact multilingual model (118M parameters) trained with contrastive objectives for semantic similarity and paraphrase detection tasks [17], optimized through contrastive fine-tuning for classification tasks.
- **Qwen3-Embedding-0.6B**: Latest generation large-scale contrastive model (600M parameters) leveraging advanced contrastive learning techniques [18], representing the first exploration of large-scale contrastive architectures for sarcasm detection tasks.

This systematic categorization enables comparative analysis between traditional BERT architectures and contrastive learning approaches. BERT models provide strong bidirectional context understanding with moderate parameter counts (110-279M), while contrastive models leverage discriminative learning to optimize representations through positive and negative sample relationships with varying scales from compact (118M parameters) to large-scale (600M parameters) architectures. The selection facilitates investigation of whether Bengali sarcasm detection benefits more from contextual language modeling (BERT) or contrastive learning approaches, and how model scale affects performance across different classification granularities.

D. Custom Model Architecture Implementation

We designed a unified framework to standardize fine-tuning across transformer models, preserving each model's unique features. All encoders are paired with a classification head.

Architecture: Tokenized Bengali text is processed by the encoder, with the [CLS] token (or first token for contrastive models) used for classification. The head applies dropout (p=0.1), a linear layer, and softmax.

Training: Cross-entropy loss is used for end-to-end optimization. Dense embeddings from contrastive models are integrated with the same head for fair comparison.

The workflow is shown in Figure 4.

E. Implementation Details and Training Strategy

All experiments were conducted using PyTorch with the transformers library 4.21.0 on a NVIDIA RTX 5080 GPU with 64GB memory and an AMD Ryzen 9 7900X processor. Training hyperparameters were standardized across all models: AdamW optimizer with learning rate 2e-5, weight decay 0.01, batch size 16, maximum sequence length 512 tokens, and early stopping with patience of 15 epochs based on validation loss to prevent overfitting.

F. Evaluation Framework and Metrics

All reported results represent performance on the held-out test set (1,241 samples) to ensure unbiased evaluation.

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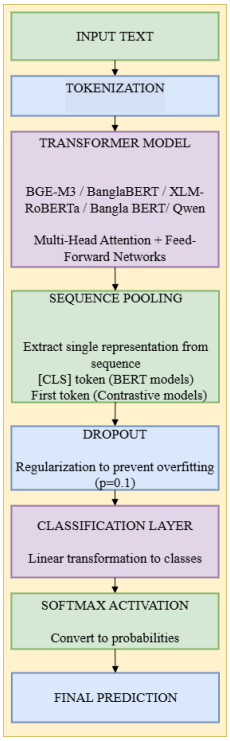


Fig. 4. Custom model architecture for transformer-based Bengali sarcasm classification.

Comprehensive evaluation employed multiple metrics to assess classification performance: accuracy, macro-averaged precision, macro-averaged recall, and macro-averaged F1-score calculated across all classes to ensure balanced evaluation regardless of class distribution.

For the hierarchical approach, we report both stage-specific performance and overall end-to-end accuracy. Macro-averaged metrics provide an unbiased performance assessment by treating all classes equally, which is particularly important given the imbalanced nature of thematic sarcasm categories.

V. Results and Analysis

A. Five-Class Direct Classification Results

Table II presents the best performance of six transformer models on direct five-class classification. BanglaBERT achieves the highest accuracy (76.15%) after 12 epochs with balanced precision (76.87%) and F1-score (74.71%), outperforming both multilingual models like XLM-RoBERTa (75.42%) and specialized contrastive models like BGE-M3 Embedding (75.75%).

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Bangla BERT(base)	71.15%	70.69%	68.30%	69.37%
BGE-M3 Embedding	75.75%	75.36%	74.06%	74.32%
Qwen3-Embedding-0.6B	69.14%	70.85%	63.77%	66.70%
XLM-RoBERTa	75.42%	75.20%	73.99%	74.19%
BanglaBERT	**76.15%**	**76.87%**	**73.69%**	**74.71%**
MiniLM-L12-v2	73.82%	73.07%	70.88%	71.83%

TABLE II
FIVE-CLASS DIRECT CLASSIFICATION RESULTS

B. Hierarchical Classification Results

The hierarchical approach demonstrates specialized model effectiveness across both stages. Stage 1 (binary classification) shows BanglaBERT excelling with 84.21% accuracy, while Stage 2 (thematic classification) reveals BGE-M3 Embedding's superiority at 88.75% accuracy with rapid convergence.

1) Stage 1 - Binary Sarcasm Detection: Table III shows BanglaBERT achieving superior performance (84.21% accuracy) with balanced precision (84.25%) and F1-score (84.20%). The 11.61% performance gap between best (BanglaBERT) and worst (Qwen3-Embedding-0.6B, 72.60%) models indicates significant sensitivity for Bengali binary sarcasm detection. Bengali-specific models (BanglaBERT: 84.21%, Bangla BERT(base): 76.87%) outperform multilingual alternatives, validating language-specific pre-training importance.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Bangla BERT(base)	76.87%	76.94%	76.88%	76.86%
BanglaBERT	**84.21%**	**84.25%**	**84.20%**	**84.20%**
BGE-M3 Embedding	81.31%	81.32%	81.31%	81.31%
Qwen3-Embedding-0.6B	72.60%	72.86%	72.62%	72.54%
MiniLM-L12-v2	77.52%	77.53%	77.52%	77.52%
XLM-RoBERTa	80.50%	80.49%	80.51%	80.50%

TABLE III
STAGE 1: BINARY SARCASM DETECTION RESULTS

2) Stage 2 - Thematic Classification: Table IV demonstrates BGE-M3 Embedding's excellence (88.75% accuracy), showcasing contrastive learning effectiveness for thematic nuances.

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Bangla BERT(base)	80.87%	81.16%	76.02%	78.06%
BanglaBERT	86.33%	83.92%	85.52%	84.67%
BGE-M3 Embedding	**88.75%**	**86.28%**	**86.29%**	**87.41%**
XLM-RoBERTa	84.41%	83.97%	80.99%	82.25%
MiniLM-L12-v2	81.99%	80.25%	78.07%	79.03%
Qwen3-Embedding-0.6B	83.16%	83.29%	78.96%	80.79%

TABLE IV
STAGE 2: STANDALONE THEMATIC CLASSIFICATION RESULTS ON SARCASTIC INSTANCES

The rapid convergence compared to Stage 1's 31 epochs indicates that thematic classification becomes more tractable with pre-filtered sarcastic content. In this stage, all models were trained and evaluated exclusively on sarcastic texts, classifying them into thematic categories. Performance spans from 80.87% (Bangla BERT(base)) to 88.75% (BGE-M3), showing substantial accuracy across all models.

3) Overall Performance Comparison: Table V demonstrates the hierarchical approach's superiority with 77.13% overall accuracy versus 76.15% for direct classification, representing a 0.98% improvement. The table shows individual stage performance and the end-to-end pipeline results.

VI. Discussion

The hierarchical approach achieves 77.13% overall accuracy versus 76.15% for direct classification, demonstrating

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Approach	Best Model	Accuracy	Macro Precision	Macro Recall	Macro F1
Five-class Direct	**BanglaBERT**	**76.15%**	**76.87%**	**73.69%**	**74.71%**
Hierarchical Approach	Stage-1 (BanglaBERT)	**84.21%**	**84.25%**	**84.20%**	**84.20%**
	Stage-2 (BGE-M3 Embedding)	**88.75%**	**86.28%**	**86.29%**	**87.41%**
	End-to-End Pipeline (BanglaBERT+ BGE-M3 Embedding)	**77.13%**	**77.92%**	**74.81%**	**75.96%**
Improvement		**+0.98%**	**+1.05%**	**+1.12%**	**+1.25%**

TABLE V
PERFORMANCE COMPARISON OF HIERARCHICAL AND DIRECT FIVE-CLASS CLASSIFICATION

that specialized models outperform unified approaches for Bengali thematic sarcasm detection. BanglaBERT's superiority in binary classification (84.21%) validates the importance of language-specific pre-training, while BGE-M3 Embedding's dominance in thematic classification (88.75%) highlights contrastive learning effectiveness for fine-grained semantic distinction.

The performance gap between Stage 2 standalone (88.75%) and overall pipeline (77.13%) reflects error propagation in cascaded systems. However, the hierarchical approach's consistent improvement over direct classification validates that specialized model benefits outweigh cumulative errors. BGE-M3's rapid convergence at epoch 3 versus BanglaBERT's 31 epochs suggests thematic pattern learning becomes more tractable with pre-filtered sarcastic content, supporting hierarchical decomposition effectiveness.

VII. CONCLUSION

This research introduces BanglaSarcThematic, the first transformer-based framework for fine-grained thematic sarcasm analysis in Bengali. We contribute the largest thematically-annotated Bengali sarcasm dataset with 12,410 comments, demonstrating that hierarchical classification with specialized models outperforms direct approaches.

Our systematic evaluation reveals that BanglaBERT excels at binary sarcasm detection, achieving 84.21% accuracy, while BGE-M3 Embedding demonstrates superior thematic classification with 88.75% accuracy. The hierarchical pipeline yields 77.13% overall accuracy compared to 76.15% for direct classification, validating task decomposition strategies for complex linguistic phenomena in Bengali digital communication contexts.

This work establishes new benchmarks for Bengali sarcasm detection and provides valuable insights for low-resource language processing. The framework's effectiveness demonstrates that specialized model selection and hierarchical approaches can effectively address thematic nuances in sarcastic discourse despite limited computational resources.

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